

- Description

The purpose of this module is to determine whether three given numbers can serve as the side lengths of a valid triangle. This is based on the principle that “the sum of any two sides must be greater than the third side”.

- Specification

module TriangleValidation

var a, b, c: real //three side lengths of a triangle

process isValidTriangle(a:real, b:real, c:real) flag: boolean

functional scenario form:

a > 0 and b > 0 and c > 0 and a + b > c and a + c > b and b + c > a && flag = true, A validated triangle, the sum of any two sides is greater than the third side

||

a <= 0 && flag = false, Not a validated triangle

||

b <= 0 && flag = false, Not a validated triangle

||

c <= 0 && flag = false, Not a validated triangle

||

a + b <= c && flag = false, Not a validated triangle

||

a + c <= b && flag = false, Not a validated triangle

||

b + c <= a && flag = false, Not a validated triangle

end process

end module

- Java Code

```
package code;
public class c01_TriangleValidation {
    public static boolean isValidTriangle(double a, double b, double c) {
        boolean flag;
        if ((a > 0) && (b > 0) && (c > 0) && (a + b > c) && (a + c > b) && (b
+ c > a)) {
            flag = true;
        } else
            flag = false;
        return flag;
    }
    public static void main(String[] args) {
        // testcase-VT(vibration testing):
        double[][] testCases = {
```

```

        {1.7, 2.845, 0.99}, //a > 0, b > 0, c > 0, a + b - c > 0,
        b + c - a > 0, a + c - b <= 0
        {5.3284, 0.712, 6.9703}, //a > 0, b > 0, c > 0, a + b - c
        <= 0, b + c - a > 0, a + c - b > 0
        {4.73336, 19.0923, 15.002}, //a > 0, b > 0, c > 0, a + c
        - b > 0, a + b - c > 0, b + c - a > 0
        {0.179, 6.90, 7.779}, //a > 0, b > 0, c > 0, a + b - c <=
        0, b + c - a > 0, a + c - b > 0
        {18, 1.3413, 13.544}, //a > 0, b > 0, c > 0, a + b - c > 0
        , b + c - a <= 0, a + c - b > 0
        {-16.6171, 6.9286, 12.4}, // a <= 0
        {0.323, -1, 3.223}, // b <= 0
        {4.73336, 19.0923, -10.643}, // c <= 0
//      };
//      // testcase-FT(fuzz testing):
//      double[][] testCases = {
//          {2.07656805837478E298, -1.7976931348623157E308, 0},
//          {2.07656805837478E298, 0, 0},
//          {2.092662514630868E298, 4.9E-324, 0},
//          {7.192970191251195E307, 8.912522337530263E302, 0},
//          {7.192969784996304E307, -1.7949137296852373E308, 0},
//          {7.192969784996304E307, -1.5210464161710564E308, 0},
//          {-1.5210464161710564E308, -
//          1.5210464161710564E308, 2.2250738585072014E-308},
//          {7.192912329778232E307, 7.822391121039473E307, 1.79769313486
//          23157E308},
//      };
//      // testcase-Z3:
//      double[][] testCases = { { 3, 4, 5 }, { 0, 3, 4 }, { 3, -
//      1, 4 }, { 4, 5, 0 }, { 1, 2, 3 }, { 1, 3, 2 },
//      { 3, 1, 2 }, { 5, 5, 8 }, };
//      V-method v method
//      double[][] testCases = {
//          {2, 2, 2},
//          {5.3284, 0.712, 6.9703},
//          {4.73336, 19.0923, 15.002},
//          {0.179, 6.90, 7.779},
//          {1, 1, 1},
//          {-16.6171, 6.9286, 12.4},
//          {0.323, -1, 3.223},
//          {4.73336, 19.0923, -10.643},
//      };
//      for (double[] testCase : testCases) {

```

```

        boolean result = isValidTriangle(testCase[0], testCase[1], testCase[
2]);
        System.out.printf("isValidTriangle(%.1f, %.1f, %.1f) = %b\n", testCa
se[0], testCase[1], testCase[2],
            result);
    }
}
}

```

- Coverage report

1. Test case number: 8
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	1.7 %	14	802	816
✓ src	1.7 %	14	802	816
✓ code	1.7 %	14	802	816
✓ c01_TriangleValidation.java	100.0 %	14	0	14
✓ c01_TriangleValidation	100.0 %	14	0	14
✓ isValidTriangle(double, double, c	100.0 %	12	0	12
✓ main(String[])	100.0 %	2	0	2

3. Vibration testing instruction coverage: 98.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	1.5 %	236	15,274	15,510
✓ src	1.5 %	236	15,274	15,510
✓ code	1.5 %	236	15,274	15,510
✓ c01_TriangleValidation.java	98.7 %	236	3	239
✓ c01_TriangleValidation	98.7 %	236	3	239
✓ isValidTriangle(double, double, c	100.0 %	37	0	37
✓ main(String[])	100.0 %	199	0	199

4. Fuzz testing branch coverage: 64.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	1.1 %	9	807	816
✓ src	1.1 %	9	807	816
✓ code	1.1 %	9	807	816
✓ c01_TriangleValidation.java	64.3 %	9	5	14
✓ c01_TriangleValidation	64.3 %	9	5	14
✓ isValidTriangle(double, double, c	58.3 %	7	5	12
✓ main(String[])	100.0 %	2	0	2

5. Fuzz testing instruction coverage: 92.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	1.4 %	221	15,289	15,510
✓ src	1.4 %	221	15,289	15,510
✓ code	1.4 %	221	15,289	15,510
✓ c01_TriangleValidation.java	92.5 %	221	18	239
✓ c01_TriangleValidation	92.5 %	221	18	239
✓ isValidTriangle(double, double, c	59.5 %	22	15	37
✓ main(String[])	100.0 %	199	0	199

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.7 %	14	826	840
▼ src	1.7 %	14	826	840
▼ code	1.7 %	14	826	840
▼ c01_TriangleValidation.java	100.0 %	14	0	14
▼ c01_TriangleValidation	100.0 %	14	0	14
isValidTriangle(double, double, double)	100.0 %	12	0	12
main(String[])	100.0 %	2	0	2

7. Z3 instruction coverage: 98.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	1.1 %	236	21,850	22,086
▼ src	1.1 %	236	21,850	22,086
▼ code	1.1 %	236	21,850	22,086
▼ c01_TriangleValidation.java	98.7 %	236	3	239
▼ c01_TriangleValidation	98.7 %	236	3	239
isValidTriangle(double, double, double)	100.0 %	37	0	37
main(String[])	100.0 %	199	0	199

8. Unimproved V-method branch coverage: 85.7%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.4 %	12	828	840
▼ src	1.4 %	12	828	840
▼ code	1.4 %	12	828	840
▼ c01_TriangleValidation.java	85.7 %	12	2	14
▼ c01_TriangleValidation	85.7 %	12	2	14
isValidTriangle(double, double, double)	83.3 %	10	2	12
main(String[])	100.0 %	2	0	2

9. Unimproved V-method instruction coverage: 98.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	1.1 %	236	21,850	22,086
▼ src	1.1 %	236	21,850	22,086
▼ code	1.1 %	236	21,850	22,086
▼ c01_TriangleValidation.java	98.7 %	236	3	239
▼ c01_TriangleValidation	98.7 %	236	3	239
isValidTriangle(double, double, double)	100.0 %	37	0	37
main(String[])	100.0 %	199	0	199